

DEFENSIVE TECHNICAL DISCLOSURE

FLAXCHEW

Chewable Capsules Containing Hydrated Natural Seeds and Botanical Extracts for Behavioral Calming, Stress Relief, and Oral Habit Substitution

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PRIOR USPTO FILING

US Provisional Patent Application No. 63/897,569

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PURPOSE

This document is prepared for public defensive publication of the technical subject matter described herein, with provenance to the inventor's earlier USPTO provisional filing.

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Provenance and source integrity

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SOURCE INTEGRITY NOTE

The SHA-256 value above identifies the unmodified USPTO-rendered specification file used as the provenance anchor. Annex A is a normalized public transcription of the filed technical content: unsupported glyphs and formatting artifacts are normalized for readability, while private correspondence details are omitted. The underlying technical wording is preserved except for those formatting and privacy normalizations.

Status of this document

This is a voluntary defensive technical disclosure prepared by the inventor. It is not a granted patent, not a WIPO/PCT publication, and not a representation that any patent right has been granted. Its purpose is to make the disclosed technical subject matter publicly accessible with a traceable link to the earlier USPTO filing record.

Technical disclosure summary

The disclosed system concerns chewable edible capsules containing a core of hydrated or partially hydrated natural seeds suspended in seed mucilage and/or plant gel, surrounded by a flexible dissolvable shell. The source specification describes controlled shell softening and dissolution, a bite-through event that releases the seed core, optional botanical, oil, vitamin or mineral components, multiple product embodiments, manufacturing approaches, and methods of use directed to oral sensory engagement and nutritional supplementation.

Technical subject matter disclosed in the source specification

- Core compositions using hydrated or partially hydrated natural seeds, including flaxseed, chia, hemp, psyllium, amaranth, sesame, sunflower, and combinations.
- Flexible dissolvable shells using gelatin or plant-based film materials such as HPMC, pectin, agar, carrageenan, pullulan, modified starch, or combinations.
- Described physical ranges and process parameters for shell thickness, softening/dissolution behavior, seed fraction, hydration, partial drying, moisture and water activity.
- Optional functional additives and product variants directed to differing sensory or wellness-oriented use cases.
- Manufacturing workflows covering seed preparation, hydration, partial drying, core formation, encapsulation, stabilization and packaging.
- Claims directed to compositions, methods, delivery systems and related embodiments as set out in Annex A.

EVIDENCE / SAFETY NOTE

Descriptions in the source specification of calming, behavioral, nutritional, pediatric or other health-related effects are reproduced as part of the inventor's technical disclosure. This publication does not establish clinical validation, medical advice, regulatory approval, product safety, or efficacy.

Annex A - normalized public transcription

US Provisional Patent Application No. 63/897,569

USPTO filing date: 11 October 2025

The following annex presents the filed technical specification in a clean public transcription. Markdown artifacts, unsupported symbols and page-layout noise are normalized. The inventor name, application identity, substantive technical description, claims and source wording are retained. Street address and correspondence email are omitted from the public copy.

EXACT USPTO SOURCE SHA-256

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PUBLIC TRANSCRIPTION NOTE

The following content is a readability-normalized transcription of the inventor's filed specification. Private correspondence fields are omitted. Source claims and statements are presented as disclosure content and are not independently validated by this publication.

PROVISIONAL PATENT APPLICATION

United States Patent and Trademark Office (USPTO)

TITLE OF INVENTION

Chewable Capsules Containing Hydrated Natural Seeds and Botanical Extracts for Behavioral Calming, Stress Relief, and Oral Habit Substitution

INVENTOR INFORMATION

Inventor Name: Robert Walasik

Cross-Reference To Related Applications

None

BACKGROUND OF THE INVENTION

Field of the Invention

The invention relates to edible compositions and delivery systems for behavioral therapy and nutritional supplementation, specifically to chewable capsules containing hydrated natural seeds designed to provide sensory satisfaction, stress relief, and behavioral substitution for oral compulsive habits such as nail-biting and stress-related chewing.

Description of Related Art

Known solutions for oral fixation include: (i) bitter nail coatings that deter biting but do not satisfy oral sensory needs; (ii) non-edible stress chews (e.g., silicone/rubber devices) that lack nutritional value and palatability; and (iii) dietary supplements (oils, powders, dry capsules) intended to be swallowed without chewing, offering no structured sensory journey.

However, existing products do not provide a chewable, flexible capsule with a hydrated seed core engineered for a rewarding textural progression (softening shell -> bite-through moment -> slippery seed interaction) that simultaneously functions as a behavioral substitute and delivers nutritional benefits (e.g., omega-3, fiber). This gap is addressed by the present invention.

SUMMARY OF THE INVENTION

The invention provides chewable edible capsules comprising:

1. **A core** of hydrated or partially hydrated natural seeds (e.g., flaxseed, chia, hemp, psyllium, amaranth, sesame, sunflower, or combinations), suspended in seed mucilage and/or a plant gel; and

2. **A flexible, dissolvable outer coating** (gelatin or plant-based film such as HPMC, pectin, agar , carrageenan, pullulan, modified starch).

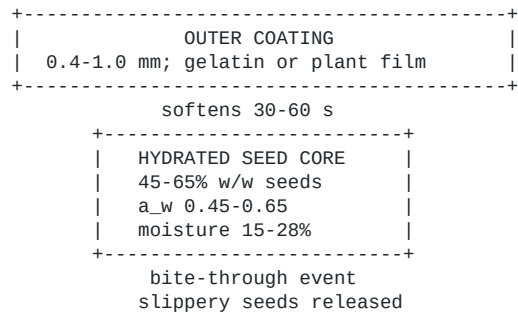
The shell is configured to soften in **30-60 seconds** and fully dissolve in **2-5 minutes** under oral conditions (~37 deg C, pH 6.5-7.5), enabling a bite-through moment that releases a slippery seed core for extended oral engagement.

Optional plant-derived oils, botanical extracts, vitamins, or minerals can be included to support calm, focus, sleep, or balance. Pediatric formats (300-500 mg; 6-8 mm diameter) are contemplated.

Primary applications include behavioral substitution for nail-biting and stress chewing, stress management, mindfulness rituals, and nutritional supplementation. Methods of manufacture and use are also provided.

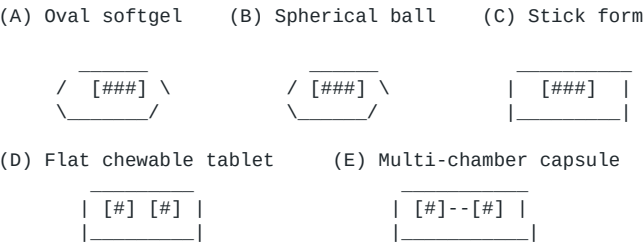
BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1: Cross-sectional View of Chewable Capsule



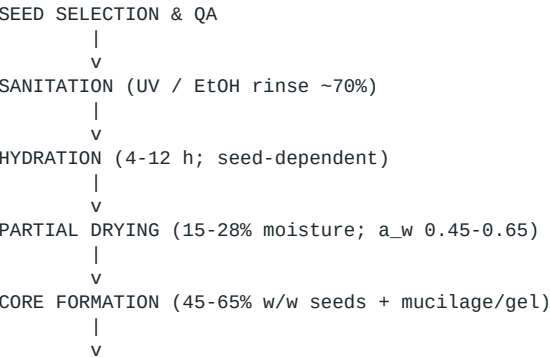
Components: (i) outer flexible coating (thickness 0.4-1.0 mm), (ii) inner hydrated seed core with seed fraction 45-65% (w/w), (iii) core water activity a_w 0.45-0.65, (iv) optional flavor/aroma micro-coating.

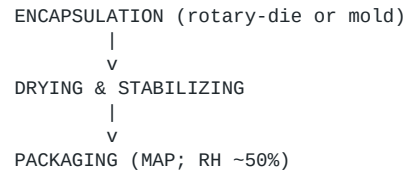
FIG. 2: Alternative Capsule Embodiments



Shows: (A) oval softgel shape, (B) spherical gummy-style ball, (C) elongated chewable stick, (D) flat chewable tablet with seed core, (E) multi-chamber capsule with layered seed types.

FIG. 3: Manufacturing Process Flowchart





Process steps: seed selection -> sanitation -> hydration (4-12h) -> partial drying (to 15-28% moisture, a_w 0.45-0.65) -> core formation (45-65% w/w seeds) -> encapsulation -> finishing/ coating -> controlled drying -> moisture-barrier packaging.

FIG. 4: Product Line Variants

VARIANT	KEY INGREDIENTS	USE
CALM	flax, chia, lavender	relax
FOCUS	hemp, amaranth, ginseng	focus
BALANCE	flax, sesame, ashwagandha	daily
SLEEP	flax, psyllium, chamomile	rest
HABITOFF	flax, neutral flavor	habit
KIDCALM	flax, fruit flavor	kids

Detailed Specifications:

- **Calm Seed(TM)**: Lavender coating (purple) - flax + chia, lemon balm, lavender
- **Focus Seed(TM)**: Citrus coating (yellow) - hemp + amaranth, ginseng, peppermint
- **Balance Seed(TM)**: Natural coating (beige) - flax + sesame, ashwagandha
- **Sleep Seed(TM)**: Blue coating - flax + psyllium, chamomile, passionflower
- **HabitOff(TM)**: Clear coating - flax primary, minimal additives, maximal texture
- **KidCalm(TM)**: Fruit-colored - pediatric format (300-500mg, 6-8mm diameter)

FIG. 5: User Experience Timeline

TIME	0-5 s	5-30 s	30-60 s	60-120 s	120 s+
STATE	Cool Smooth	Softening	Bite-through	Seed explore	Calm state
EFFECT	Attention shift	Calming anticipation	Reward response	Engagement stress relief	Resolution satisfied

Phases: Initial contact (0-5s) -> coating softening (5-30s) -> bite-through moment (30-60s) -> seed exploration/chewing (60-120s) -> calm resolution (120s+).

DETAILED DESCRIPTION OF THE INVENTION

A. Composition Components

A1. Seed Core (Hydrated Natural Seeds)

The core comprises one or more natural seeds in a hydrated or partially hydrated state.

Preferred seeds:

- Flaxseed (*Linum usitatissimum*)
- Chia (*Salvia hispanica*)
- Hemp seed (*Cannabis sativa*; non-psychoactive)
- Psyllium (*Plantago ovata*)
- Amaranth, sesame, sunflower

Hydration process:

- Soaking in potable water for 4-12 hours (seed-dependent) to achieve partial swelling and surface mucilage/gel formation
- Flax/chia: typically 6-8 hours
- Hemp/amaranth: typically 4-6 hours

Partial drying:

- Target moisture content: **15-28% (w/w)**
- Achieving water activity: **a_w 0.45-0.65** (balances mouthfeel and microbial stability)

Seed fraction in core:

- **45-65% (w/w)** seeds dispersed in seed mucilage and/or plant gel
- Plant gel options: pectin 1-3%, agar 0.3-1.0%, HPMC 1-4% in the core phase

Sensory role:

- Hydrated seeds present a slippery, chewable texture
- Optional "crack" sensation upon bite if hydration is controlled to maintain seed integrity

A2. Outer Coating (Flexible, Dissolvable Shell)

The coating is flexible and chewable, designed to:

- Soften in **30-60 seconds**
- Fully dissolve in **2-5 minutes** at ~37 deg C, pH 6.5-7.5

Materials:

- **Gelatin** (animal-derived)
- **Plant-based films:** HPMC, pectin, agar, carrageenan, pullulan, modified starch, or combinations

Coating thickness:

- **0.4-1.0 mm**, optimized for:
- Initial smoothness
- Controlled softening rate
- Satisfying bite-through resistance

Optional shell features:

- Microencapsulated aromatics (lavender, citrus, mint) - release on bite
- Natural pigments (beetroot, spirulina, turmeric) for color coding
- Flavor micro-coating (vanilla, fruit essences)

A3. Optional Functional Additives**Plant oils:**

- Hemp seed oil (omega-3/6, non-psychoactive)
- Flaxseed oil (ALA omega-3)
- Coconut/MCT oil (smooth mouthfeel)
- Evening primrose oil (GLA)

Botanical extracts (calm/focus/sleep):

- **Calming:** Lemon balm, lavender , chamomile, passionflower
- **Adaptogenic:** Ashwagandha, holy basil
- **Focus:** Ginseng, guarana, L-theanine
- **Sleep:** Valerian, melatonin (low dose)
- **Refreshing:** Mint, citrus oils

Micronutrients:

- Magnesium (e.g., glycinate)
- B-complex vitamins
- Vitamin D3, E
- Zinc

Regulatory note: The invention contemplates inclusion of non-psychoactive hemp- derived components or CBD where legally permitted. Formulations can be adapted for regional compliance.

B. Manufacturing Process**B1. Seed Preparation**

1. **Selection & QA:** Food-grade seeds; removal of foreign matter
2. **Surface sanitation** (food-grade):
 - Aqueous ethanol rinse -> **70% v/v** followed by drying
 - Optional UV-C exposure for surface microbial reduction
 - Optional mild thermal treatment (<60 deg C; time sufficient to reduce total plate count)
3. **Hydration/Soaking:**
 - Duration: 4-12 hours (seed-specific)
 - Promotes swelling and mucilage formation
4. **Partial drying:**
 - Method: Vacuum drying or low-temperature air drying (<40 deg C)
 - Target: **15-28% moisture (w/w)**
 - Achieving: **a_w 0.45-0.65**

B2. Core Formation

- Combine hydrated seeds with plant gel (pectin/agar/HPMC) to desired dose
- Core mass range: 300-1000 mg
- Optionally incorporate plant oils/botanicals
- Target seed fraction: **45-65% (w/w)**

B3. Encapsulation**Rotary-die softgel/HPMC process** (preferred for scale):

- Two films formed simultaneously

- Core material injected between films
- Seam sealed and cut to shape

Alternative (prototyping):

- Pre-forming shells in molds (silicone)
- Filling with core material
- Sealing with secondary film layer

B4. Finishing & Stabilization**Coating enhancement:**

- Optional flavor/aroma micro-coating
- Natural color pigments

Controlled drying:

- Achieve target shell properties (softening/dissolution profile)
- Controlled humidity environment

Stabilization:

- Antioxidants: Mixed tocopherols **0.02-0.1% (w/w)**, rosemary extract
- Purpose: Improve oxidative stability of seed oils

Packaging:

- Moisture-barrier blisters or pouches
- Optional modified atmosphere packaging (MAP)
- Storage humidity: **RH -> 50%**
- Target shelf life: **12-24 months** at ambient conditions

C. Product Embodiments (Variants)**Variant 1: Calm Seed(TM) (Relaxation Focus)**

- **Seeds:** Flax + chia blend
- **Botanicals:** Lemon balm, lavender , chamomile
- **Additives:** Magnesium glycinate, L-theanine
- **Flavor:** Vanilla-lavender
- **Coating color:** Soft purple/lavender
- **Application:** Evening stress relief, pre-sleep ritual

Variant 2: Focus Seed(TM) (Mental Clarity)

- **Seeds:** Hemp + amaranth
- **Botanicals:** Ginseng, citrus oils, peppermint
- **Additives:** B-complex vitamins
- **Flavor:** Citrus-mint
- **Coating color:** Yellow/lime

- **Application:** Work/study focus, afternoon energy

Variant 3: Balance Seed(TM) (Emotional Stability)

- **Seeds:** Flax + sesame
- **Botanicals:** Ashwagandha, holy basil
- **Additives:** Omega-3 oil, zinc
- **Flavor:** Natural/caramel
- **Coating color:** Beige/golden
- **Application:** Daily emotional regulation

Variant 4: Sleep Seed(TM) (Bedtime Support)

- **Seeds:** Flax + psyllium
- **Botanicals:** Chamomile, passionflower
- **Additives:** Magnesium, melatonin (0.5-1mg)
- **Flavor:** Coconut-vanilla
- **Coating color:** Soft blue
- **Application:** Pre-bedtime wind-down

Variant 5: HabitOff(TM) (Behavioral Substitution)

- **Seeds:** Flaxseed primary
- **Botanicals:** Minimal (or mild mint)
- **Focus:** Maximum texture satisfaction for oral fixation
- **Flavor:** Minimal/natural
- **Coating color:** Clear or neutral
- **Application:** Nail-biting/skin-picking substitute

Variant 6: KidCalm(TM) (Pediatric Version)

- **Seeds:** Flaxseed (smaller capsule format)
- **Size:** Unit mass **300-500 mg**, diameter **6-8 mm**
- **Botanicals:** Mild chamomile, fruit-derived flavonoids (no stimulants)
- **Additives:** Vitamin D, omega-3 (age-appropriate doses)
- **Flavors:** Natural fruit (strawberry, orange, apple)
- **Coating color:** Bright, appealing (pink, orange, green)
- **Safety:** Age-appropriate size, reduced choking hazard, allergen-free options
- **Application:** Emotional regulation training (ages 5-12), ADHD support, stress management for children

D. Sensory Experience Design (User Journey)

Phase 1 (0-5 seconds): Initial Contact

- Capsule placed on tongue
- Cool, smooth surface texture

- User anticipation begins
- **Psychological effect:** Focus shift, distraction from stress trigger

Phase 2 (5-30 seconds): Softening

- Coating begins to dissolve
- Flavor release (if flavored)
- Gentle aroma (if microencapsulated)
- **Psychological effect:** Calming anticipation, mindfulness moment

Phase 3 (30-60 seconds): The Bite-Through

- User bites through softened coating
- Satisfying "crack" or "snap" sensation
- Seed core released into mouth
- **Psychological effect:** Reward response, dopamine micro-burst

Phase 4 (60-120 seconds): Seed Interaction

- Slippery, gel-coated seeds explored with tongue
- Individual seeds optionally cracked between teeth
- Mild nutty/earthy flavor profile
- **Psychological effect:** Continued oral engagement, stress dissipation

Phase 5 (120+ seconds): Resolution

- Seeds fully chewed or swallowed
- Residual flavor/texture sensation
- Calm, satisfied state achieved
- **Psychological effect:** Habit loop completed, stress cycle broken

Neurochemical mechanism:

- Oral stimulation -> sensory cortex activation
- Anticipation phase -> dopamine precursor release
- Bite satisfaction -> dopamine reward signal
- Nutritional delivery -> sustained serotonin support (via omega-3, tryptophan in seeds)

This creates a **"functional gratification loop"** - a healthy behavioral cycle that replaces destructive oral habits.

E. Applications and Use Cases

Therapeutic/Behavioral Applications:

1. **Onychophagia** (nail-biting) - oral substitute reduces urge frequency
2. **Dermatillomania** (skin-picking) - redirects compulsive behavior
3. **Trichotillomania** (hair-pulling) - provides alternative oral stimulation
4. **ADHD/Autism** - sensory stimulation tool, focus aid
5. **Anxiety disorders** - grounding technique, mindfulness anchor

6. **Smoking cessation** - oral fixation substitute during quit programs

Wellness/Lifestyle Applications:

1. **Stress management** - office workers, students, high-stress professions
2. **Mindfulness practice** - meditation aid, breathing exercise companion
3. **Pre-sleep ritual** - wind-down routine
4. **Travel anxiety** - portable calm tool (flights, long drives)
5. **Performance stress** - exam preparation, public speaking

Pediatric Applications:

1. **Emotional regulation training** (ages 5-12)
2. **Classroom focus tool** - quiet, non-disruptive sensory aid
3. **Tantrum prevention** - redirection technique for caregivers
4. **Bedtime routine** - calming pre-sleep ritual

CLAIMS

Claim 1 (Independent - Composition)

A chewable edible composition comprising:

- (a) a core with hydrated or partially hydrated natural seeds selected from flaxseed, chia, hemp, psyllium, amaranth, sesame, sunflower, or combinations thereof, the seeds having a moisture content of 15-28% (w/w) and a seed fraction of 45-65% (w/w) within a mucilage and/or plant-gel matrix; and
- (b) a flexible, dissolvable outer coating comprising gelatin, hydroxypropyl methylcellulose (HPMC), pectin, agar, carrageenan, pullulan, modified starch, or combinations thereof, the coating having a thickness of 0.4-1.0 mm and being configured to soften in 30-60 s and dissolve within 2-5 min under oral conditions;

wherein, upon softening and bite-through, the composition releases a slippery seed core providing extended oral engagement and behavioral substitution for stress-related oral habits.

Claim 2 (Method of Use - Behavioral)

A method of providing behavioral substitution for stress-related oral fixation, comprising administering the composition of claim 1 to a subject and allowing softening and bite-through to engage oral sensory feedback in lieu of nail-biting or stress chewing.

Claim 3 (Manufacturing Method)

A method of producing the composition of claim 1 comprising: soaking seeds for 4-12 h; partially drying to 15-28% moisture and a_w 0.45-0.65; forming a core with 45-65% seeds (w/w) in seed mucilage and/or plant gel; and encapsulating the core in a flexible, dissolvable coating of claim 1 under controlled humidity.

Claim 4 (Botanical Option)

The composition of claim 1 further comprising plant-derived oils and/or botanical extracts selected from hemp seed oil, flaxseed oil, coconut or MCT oil, lemon balm, lavender, chamomile, ashwagandha, L-theanine, mint, or combinations thereof.

Claim 5 (Pediatric Format)

The composition of claim 1 in a pediatric format having a unit mass of 300-500 mg and diameter 6-8 mm, with fruit flavors and plant-only ingredients.

Claim 6 (Vegan Shell)

The composition of claim 1 wherein the coating is plant-based, comprising HPMC, pectin, agar, carrageenan, pullulan, or combinations thereof.

Claim 7 (Gelatin Shell)

The composition of claim 1 wherein the coating comprises gelatin.

Claim 8 (Aroma Release)

The composition of claim 1 wherein the coating comprises microencapsulated aromatic compounds configured to release upon bite-through.

Claim 9 (CBD Where Legal)

The composition of claim 1 optionally comprising non-psychoactive hemp-derived components or CBD, where legally permitted, to enhance calming effects.

Claim 10 (Smoking Cessation Use)

The method of claim 2 wherein the composition is used as an oral substitute during smoking cessation.

Claim 11 (Shelf Stability)

The composition of claim 1 packaged in moisture-barrier primary packaging and stabilized to a shelf life of 12-24 months at ambient conditions.

ABSTRACT

A chewable edible capsule is disclosed, comprising a core of hydrated or partially hydrated natural seeds (e.g., flax, chia, hemp) suspended in a plant gel, and a flexible, dissolvable outer coating (gelatin or plant-based). The coating softens in 30-60 s and dissolves within 2-5 min under oral conditions, enabling bite-through to release a slippery seed core that promotes extended oral engagement. The composition functions as a behavioral substitute for stress-related oral habits (nail-biting, stress chewing) while delivering nutritional benefits (omega-3, fiber, botanical extracts). Methods of manufacture and use are provided. Pediatric formats and variants for calm, focus, sleep, and behavioral applications are contemplated.

ADVANTAGES (Concise)

1. **Novel modality:** First hydrated seed bite-through capsule specifically designed for behavioral substitution
2. **Triple benefit:** Behavioral (oral fixation substitute) + sensory (rewarding texture) + nutritional (omega-3, fiber, botanicals)
3. **Natural & safe:** Food-grade, plant-based ingredients; biodegradable materials
4. **Customizable platform:** Modular design allows infinite seed/coating/additive combinations
5. **Wide demographic:** Adults (stress management), children (emotional regulation), elderly (mindfulness)
6. **Regulatory flexible:** Core formulation GRAS-compliant; optional ingredients region-specific

7. **Scalable manufacturing:** Compatible with existing softgel/capsule equipment
8. **Broad patent coverage:** Claims cover seeds, coatings, processes, applications - difficult to circumvent
9. **Market differentiation:** Creates new category ("Behavioral Nutrition Chews") with no direct competition

INVENTOR DECLARATION

I hereby declare that I am the original inventor of the subject matter disclosed herein, and that this invention has not been previously published, publicly disclosed, or patented by any other party to my knowledge.

Inventor Name: Robert Walasik

Inventor Signature: /Robert Walasik/

Date: 11th of October 2025

END OF PROVISIONAL PATENT APPLICATION

Document Prepared By: Robert Walasik with AI Patent Advisory

Total Word Count: ~4,100 words (optimized from original 5,200)

Total Claims: 11 (tight & broad with technical parameters)

Figures: 5 (with ASCII diagrams included)

Technical Parameters Included:

- Water activity (a_w): 0.45-0.65
- Moisture content: 15-28% (w/w)
- Seed fraction: 45-65% (w/w)
- Coating thickness: 0.4-1.0 mm
- Softening time: 30-60 seconds
- Dissolution time: 2-5 minutes
- Pediatric size: 300-500 mg, 6-8 mm diameter
- Shelf life: 12-24 months
- Storage RH: 50%
- Antioxidant range: 0.02-0.1% (w/w)